

Generation Interconnection System Impact Study Report

For Queue Project Q14-0342

Asheville 138 kV

April 19, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0342. The Interconnection Customer proposes a 295.7 MW natural gas combined cycle facility interconnecting at the Asheville 138 kV substation in the Virginia area. The request was received on June 15, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 165 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,935,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: High Ridge to Jerome 138 KV
Contingency: Taber to Dixleta 345 KV
Loading: 125.7% (Rating: 601 MVA)
DFAX: 6.9%
Pre-project loading: 122.3%
2. Facility: Taber to Glenannan 345 KV
Contingency: Sierra Pacific Lumber 138/1 kV XFMR
Loading: 125.0% (Rating: 1,656 MVA)
DFAX: 16.6%
Pre-project loading: 122.1%
3. Facility: Taber to Dixleta 345 KV line
Contingency: Sierra Pacific Lumber 138 kV/1 kV Auto
Loading: 121.3% (Rating: 1,619 MVA)
DFAX: 15.0%
Pre-project loading: 118.5%
4. Facility: Sunbright to Fairland 138 KV line
Contingency: Taber to Glenannan 345 KV
Loading: 113.3% (Rating: 673 MVA)
DFAX: 6.1%
Pre-project loading: 110.6%
5. Facility: Sierra Pacific Lumber 138 kV/1 kV Auto
Contingency: Glenannan to Taber 345 KV
Loading: 111.9% (Rating: 963 MVA)
DFAX: 5.8%
Pre-project loading: 110.1%
6. Facility: Sierra Pacific Lumber 345/1kV XFMR
Contingency: Glenannan to Taber Substation 345 KV
Loading: 111.9% (Rating: 963 MVA)

DFAX: 5.8%

Pre-project loading: 110.1%

7. Facility: Sunbright to Colesville 138 KV

Contingency: Taber to Glenannan 345 KV Line

Loading: 111.1% (Rating: 736 MVA)

DFAX: 6.1%

Pre-project loading: 108.6%

8. Facility: Dixleta Substation to Drum 138 KV

Contingency: Taber to Glenannan 345 KV

Loading: 109.8% (Rating: 648 MVA)

DFAX: 5.1%

Pre-project loading: 107.4%

9. Facility: Dixleta to Hughson 345 KV

Contingency: Dixleta to Rustic 345 KV Line

Loading: 107.8% (Rating: 1,350 MVA)

DFAX: 16.7%

Pre-project loading: 104.1%

10. Facility: Rustic to Dixleta Substation 345 KV

Contingency: Dixleta to Hughson Substation 345 KV L

Loading: 107.4% (Rating: 1,314 MVA)

DFAX: 15.3%

Pre-project loading: 104.0%

11. Facility: Louisa Generation Facility to Carrabelle 345 KV

Contingency: North Anna to Louisa Generation Facility 345 KV

Loading: 106.1% (Rating: 1,562 MVA)

DFAX: 10.6%

Pre-project loading: 104.1%

12. Facility: Tombstone to Marcus Hook Energy Lp 500 KV

Contingency: Curecanti to Eliza Creek 345 KV L

Loading: 105.4% (Rating: 2,341 MVA)

DFAX: 6.2%

Pre-project loading: 104.7%

13. Facility: Rustic to Hughson 345 KV L

Contingency: Rustic to Dixleta 345 KV

Loading: 103.9% (Rating: 1,268 MVA)

DFAX: 11.6%

Pre-project loading: 101.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
High Ridge to Jerome 138 KV	Rebuild line with higher-capacity conductor	\$2,411,000	6% (\$154,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Taber to Glenannan 345 KV L	Rebuild line with higher-capacity conductor	\$21,615,000	6% (\$1,232,000)
Taber to Dixleta 345 KV Line	Rebuild line with higher-capacity conductor	\$21,099,000	6% (\$1,201,000)
Sunbright to Fairland 138 KV line	Rebuild line with higher-capacity conductor	\$538,000	6% (\$34,000)
Sierra Pacific Lumber 138 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$6,581,000	7% (\$432,000)
Sierra Pacific Lumber 345-1 kV TX	Replace transformer with higher-rated unit	\$18,058,000	7% (\$1,185,000)
Sunbright Substation to Colesville 138 KV	Rebuild line with higher-capacity conductor	\$1,489,000	6% (\$94,000)
Dixleta Substation to Drum 138 KV	Reconductor line	\$540,000	7% (\$36,000)
Hughson to Dixleta 345 KV	Reconductor line	\$12,634,000	5% (\$688,000)
Rustic to Dixleta Substation 345 KV	Reconductor line	\$23,284,000	5% (\$1,261,000)
Louisa Generation Facility to Carrabelle 345 KV	Reconductor line	\$28,841,000	5% (\$1,521,000)
Tombstone to Marcus Hook Energy Lp 500 KV	Reconductor line	\$40,134,000	5% (\$2,163,000)
Rustic to Hughson 345 KV L	Reconductor line	\$16,839,000	6% (\$934,000)
	Total allocated to project		\$10,935,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.